

DENNIS FARMER

Ann Arbor, Michigan

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Education

University of Michigan - Ann Arbor, MI

Anticipated May 2027

B.S.E. Data Science - College of Engineering, Minor in Music

GPA: 3.87/4.00

Current Coursework: Web Systems, Regression Analysis, Music and AI, Digital Sound Synthesis

Coursework: Machine Learning, Computational Methods for Statistics, Data Structures and Algorithms,
Probability Theory, Linear Algebra, Discrete Mathematics

Projects

Shazam Music Recognition Clone - Project Lead 🔗

June 2025 – Present

Michigan Data Science Team

Ann Arbor, MI

- Replicated the Shazam algorithm in NumPy and integrated it into an Expo app interface, with a web-scraped song library stored using MySQL, in collaboration with one other project co-lead
- Designing and teaching an 8-week curriculum to teach project members the short-time Fourier transform, the Shazam algorithm, SQL, and React Native, for members to create their own version of Shazam

Music Recommender System 🔗

March 2025 – May 2025

Personal Project

Ann Arbor, MI

- Built an automatic playlist continuation model with a web interface, trained on a web-scraped dataset of 118,000 audio tracks along with user interaction data (Beautiful Soup, scikit-learn, SQL, Streamlit, Plotly)
- Created API to extract audio embeddings using containerized CNN model trained on genre classification (Tensorflow, AWS Lambda, Docker, Flask, ngrok)
- Implemented kernel principle component analysis on audio embeddings, as well as latent matrix factorization methods, for track recommendation via content-based and collaborative filtering respectively

Convolutional Neural Network for Image Classification 🔗

September 2023 – December 2023

Michigan Data Science Team

Ann Arbor, MI

- Designed a convolutional neural network architecture to distinguish between AI-generated faces and real human faces with a team of five people (PyTorch)
- Wrote a web application for uploading images and receiving classification results via API, with interpretability via GradCam convolutional layer visualizations (React, Express)

Research Experience

Machine Learning Researcher

May 2025 - Present

Cooperative Institute for Great Lakes Research

Ann Arbor, MI

- Utilizing PyTorch and HPC to recommend ideal buoy sensor placements to enhance weather forecasting
- Training convolutional gaussian neural processes on multi-modal environmental Great Lakes data for statistical downscaling and surface temperature uncertainty quantification (PyTorch)

Roaming Behavior of Domestic Cats 🔗

October 2024 - December 2024

Class Research Paper - Computational Methods for Statistics

Ann Arbor, MI

- Developed a beta regression model based on demographic variables to predict proportion of time domestic cats spent away from home, with negligible bias and MSE of 4.052 verified with Monte Carlo simulations
- Concluded cats roam farther away in dry conditions than during rainfall with a linear mixed-effects model, with bootstrapped 95% confidence interval for difference in means of (0.276 km, 1.069 km)

Causal Discovery - Philosophy of Science 🔗

September 2022 – August 2023

University of Michigan - Dr. Patrick Grim

Ann Arbor, MI

- Researched genetic algorithms for causal Bayesian networks based on Judea Pearl's theory of causality, and ideas from causal discovery literature, to model adaptive causal structures of scientific theories (Python)

Skills

Languages: C++, Python (PyTorch, Pandas, Numpy, Beautiful Soup, Flask), R, SQL, Bash, Faust, ChuckK

Tools and Software: Git, Docker, ngrok, Vim, Make, VS Code, GNU Debugger (gdb), Valgrind